

Heim's Mass Formula (1982/1989)

Consolidated Report

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Abstract

This document presents a consolidated version of Burkhard Heim's mass formula for elementary particles, incorporating the corrections and extensions published in 1989 into the original 1982 formulation. The unified theory describes the masses, lifetimes, fine-structure constant, and neutrino states of elementary particles based on a geometric-dynamical structure of spacetime.

Contents

1 Introduction

After DESY physicists programmed and calculated the mass formula published in *Elementarstrukturen der Materie* (Heim 1984), the formula was extended in 1989. A 57-page report with a new formula and calculation results was sent to MBB/DASA, but the code was later lost. Parts of the formulae were reprogrammed by the research group "Heim Theory" (Dr. A. Müller). Missing brackets in long equations were corrected. The 1989 version includes lifetimes of basic states, neutrino masses, and the fine-structure constant, which are presented here alongside the corrected 1982 foundation.

2 Mass of Basic States and of the Excited States of Elementary Particles

2.1 Quantum Numbers and Invariants (1982 with 1989 corrections)

Symbols:

- k : configuration number ($k = 0$: no ponderable particle). For ponderable corpuscles only $k = 1$ and $k = 2$ are allowed.
- ε : time helicity ($\varepsilon = \pm 1$).
- G : number of quasi-corpuscular internal structural constituents, $G = k + 1$.

- B : baryon number, $B = k - 1$.
- P : double isospin, $P = 2s$.
- I : number of components in an isospin multiplet, $I = P + 1$, $0 \leq P \leq G$.
- Q : double space spin, $Q = 2J$.
- $K(\lambda)$: doublet number ($K(\lambda) = 0$ or 1), $1 \leq \lambda \leq \Lambda = 4 - k$.
- C : structure distributor (strangeness), corrected in 1989:

$$C = 2(P\varepsilon_P + Q\varepsilon_Q)(k - 1 + K)/(1 + K)/k$$

- q_x : electric charge quantum number with sign for component x of the isospin multiplet.
- $q = |q_x|$.

The angles defining time helicity are:

$$\alpha_P = \pi Q \left(K + \binom{P}{2} \right), \quad \alpha_Q = \pi Q \left[Q + \binom{P}{2} \right] \quad (1989 \text{ correction}).$$

The charge quantum number (1989 correction) is:

$$q_x = \frac{1}{2} [(P - 2x + 2)[1 - KQ(2 - k)] + \varepsilon[k - 1 - (1 + K)Q(2 - k)] + C].$$

2.2 Mass Formula (1982 foundation with 1989 updates)

The mass formula for elementary particles is built from:

$$M = \mu\alpha_+ [(G + S + F + \Phi) + 4q\alpha_-].$$

The parts G and S are the same as $\underline{G}$ and K in the 1982 equation (XII) (using n, m, p instead of n_1, n_2, n_3). μ is the mass element as in eq. (VI). The constants $\alpha_{\pm}$ are:

$$\alpha_+ = \frac{\sqrt[6]{\eta}}{\eta^2} = 1 - \vartheta \frac{2(1 - \sqrt{\eta})}{3(1 + \sqrt{\eta})} = \sqrt{2\eta} - 1, \quad \alpha_- = (\alpha_+ + 1)\eta - 1.$$

The abbreviations for F and Φ (depending on quantum numbers) are:

$$\begin{aligned} F &= 2nQ_n [1 + 3(n + Q_n + nQ_n) + 2(n^2 + Q_n^2)] \\ &\quad + 6mQ_m(1 + m + Q_m)N_2 + 2pQ_pN_3 + \phi(p, \sigma) \cdot \delta(N), \\ \Phi &= P(-1)^{P+Q}(P + Q)N_5 + Q(P + 1)N_6, \\ \phi &= \phi(p, \sigma), \quad \delta(0) = 1 \quad (\delta(N) = 0 \text{ for } N \neq 0). \end{aligned}$$

With:

$$\begin{aligned}
\phi &= \frac{N_4 p^2}{1+p^2} \frac{\sigma + Q_\sigma}{\sqrt{1+\sigma^2}} \left(\sqrt[4]{2} - 4BUW_{N=0}^{-1} \right) \\
&+ P(P-2)^2 \left(1 + \frac{K(1-q)}{2\alpha\vartheta} \right) \left(\frac{\pi}{e} \right)^2 \sqrt{\eta_{12}} (Q_m - Q_n) \\
&- (P+1) \binom{Q}{3} / \alpha, \\
U &= 2^Z \left[P^2 + \frac{3}{2}(P-Q) + P(1-q) + \frac{4\kappa B(1-Q)}{3-2q} \right. \\
&\quad \left. + (k-1) \left\{ P + 2Q - \frac{4\pi(P-Q)(1-q)}{\sqrt[4]{2}} \right\} \right] \eta_{qk}^{-2}, \\
Z &= k + P + Q + K \quad (\text{compare with eq. B51}).
\end{aligned}$$

The functions Q_i from eq. (X) remain unchanged. For n_1, n_2, n_3, n_4 we write n, m, p, σ . The constants $\eta_{q,k}, \vartheta, \eta$ (with $\eta_{10} = \eta, \vartheta_{1,0} = \vartheta$), and the functions N_1, N_2 are as in eq. (IX). The remaining N_i ($i > 2$) are:

$$\begin{aligned}
\ln(N_3 k/2) &= (k-1) \left[1 - \pi \frac{1 - \eta_{q,k}}{1 + \sqrt{\eta_{q,1}}} \left\{ 1 - u \frac{\eta_{q,1}}{\vartheta_{q,1}} \left(1 - \frac{\alpha_-}{\alpha_+} \right) (1 - \sqrt{\eta})^2 \right\} \right] \\
&\quad - \frac{2}{3\pi e} (1 - \sqrt{\eta})^2 \left(6\pi^2 e^2 / \vartheta \frac{1 + \sqrt{\eta_{q,1}}}{1 - \eta} - 1 \right), \\
N_4 &= \frac{4}{k} [1 + q(k-1)], \\
N_5 &= A \left[1 + k(k-1) 2^{k^2+3} N(k) A \frac{1 - \sqrt{\eta_{q,k}}}{1 + \sqrt{\eta_{q,k}}} \right]^2, \\
A &= \frac{8}{\eta} \left(1 - \frac{\alpha}{\alpha_+} \right) \left(1 - \frac{3\eta}{4} \right), \\
N(k) &= Q_n + Q_m + Q_p + Q_\sigma + k(-1)^k 2^{k^2-1}, \\
N_6 &= \frac{2k}{\pi e \vartheta} \left[\sqrt{k}(k^2-1) \frac{N(k)}{\sqrt{\eta_{1,k}}} \left\{ q - (1-q) \frac{N'(k)}{Q_n \sqrt{\eta_{1,k}}} \right\} + (-1)^{k+1} \right] \\
&\quad \cdot \eta \left(1 - \frac{\alpha_-}{\alpha_+} \right) \left(4 \frac{1 - \sqrt{\eta}}{1 + \sqrt{\eta}} \right)^2 Q_\sigma, \\
N'(k) &= Q_n + Q_m + Q_p + Q_\sigma - 2k - 1.
\end{aligned}$$

2.3 Limits of Structure Zone Fillings

Let L be the upper barrier such that when reached, the filling of zone x disappears and the next higher zone is increased by 1. With $L_{(x)}(x-1)$ for this barrier and $M_0 = M(N=0)$,

the limits are:

$$\begin{aligned}
-Q_n &\leq n \leq L_{(n)} = \frac{\sqrt[3]{(P+1)M_0}}{2\mu\alpha_+N_1} - Q_n, \\
-Q_m &\leq m \leq L_{(m)}(n), \quad \text{with} \\
2(Q_m + L_{(m)}(n))^3 + 3(Q_m + L_{(m)}(n))^2 + Q_m + L_{(m)}(n) &= 4N_1(n + Q_n)^3/N_2, \\
-Q_p &\leq p \leq L_{(p)}(m), \quad \text{with} \\
2L_{(p)}(m) &= \sqrt{24\frac{N_2}{N_3}(m + Q_m)^2 + 1} - 2Q_p - 1, \\
-Q_\sigma &\leq \sigma \leq L_{(\sigma)}(p), \quad \text{with} \\
2L_{(\sigma)}(p) &= N_3(p + Q_p) - 2Q_\sigma.
\end{aligned}$$

2.4 Selection Rule and Excitation Function

The selection rule expressing n, m, p, σ by k, P, Q, K, q, N is described by eq. (XXIX). For $N > 0$, $f(N)$ is the excitation function. For $W_{\nu x} \equiv W_{N=0}$ (independent of exciting state):

$$W_{N=0} = Ae^x(1 - \eta)^L + (P - Q) \left(1 - \binom{P}{2}\right) \left(1 - \binom{Q}{3}\right) (1 - \sqrt{\eta})^2 \sqrt{2},$$

with

$$\begin{aligned}
A &= 8gH[2 - k + 8H(k - 1)]^{-1}, \\
H &= Q_n + Q_m + Q_p + Q_\sigma, \\
g &= Q_n^2 + Q_m^2 + \left(\frac{Q_p^2}{k}\right) e^{k-1} + \exp\left(\frac{1 - 2k}{3}\right) - H(k - 1), \\
L &= (1 - k)Q(2 - k), \\
x &= \left[1 - Q - \binom{P}{2}\right] (2 - k) + \frac{1}{4B} \left[a_1 + \frac{k^3}{4H} \left(a_2 + \frac{a_3}{4B}\right)\right], \\
B &= 3H[k^2(2k - 1)]^{-1}.
\end{aligned}$$

The combinatorial relations for a_1, a_2, a_3 are given in the 1989 document (eqs. B29-B31).

The excitation function is:

$$f(N) = aN/(N + 1) + bN,$$

with substitutions (1989):

$$\begin{aligned}
a &= \frac{P^2}{kX\eta_{q,k}^2\sqrt{\eta_{q,k}}}(1 - k/4) + (k - 1) \left\{ \frac{\pi}{4} \binom{P}{3} - \eta_{1,1}\eta_{1,2} \binom{P}{2} \right\}, \\
X &= k \left[4\alpha \frac{(B + k + 1)(1 + 5\alpha^2)}{(1 - \alpha^2)(1 - 5\alpha^2)} - 2\frac{3\alpha}{4\pi} \right]^2 - q \left[\frac{\pi}{2} - 1 - \alpha^2 \frac{\pi e(1 + \sqrt{\eta})}{2\vartheta(1 - 6\alpha^2)} \right] + 1, \\
b &= \frac{1}{2\eta_{qk}^2\sqrt{\eta_{qk}}} \left[\frac{\alpha\vartheta}{8}(P^2 + 1) \left[\frac{1}{2}(1 + \sqrt{\eta}) \left(1 + \eta_{1,1}\eta_{1,2} \frac{3}{4} \binom{P}{3} (k - 1) \right) \right] \right. \\
&\quad \left. + (k - 1) \left\{ \frac{\vartheta_{1,2}}{\vartheta} - 8 \binom{P}{2} (P^2 + 1)^{-1} \right\} \right] - C, \\
C &= \pi(1 - \sqrt{\eta})^2 \left[1 + \sqrt{\pi}(k - 1) + \frac{P}{k^3} \left(\frac{3}{e} + q(8 + \eta_{qk}) + \frac{4\pi e}{\sqrt{\eta}}(1 - k) \left[1 - q \frac{3\pi\eta}{5e\eta_{qk}} \right] \right. \right. \\
&\quad \left. \left. - 2(k - 1) \binom{P}{2} (3 - P) \left\{ 2e(\eta + \eta_{qk}) + \frac{\varepsilon q_x \pi e}{3\sqrt{\eta}} \right\} + \frac{8\pi e k(k - 1)}{\sqrt{\eta}} \left[\frac{e}{\sqrt{\eta}} - \frac{q}{\sqrt{e}} \right] \right) \right] \\
&\quad + \left(\frac{2eKq}{\eta^2} \right) (2 - k)(1 - \eta)^2.
\end{aligned}$$

2.5 Numerical Determination of Quantum Numbers

Given k, P, Q, K, q_x , and excitation N , the right-hand side of eq. (XXXV) is:

$$(n + Q_n)^3 \alpha_1 + (m + Q_m)^2 \alpha_2 + (p + Q_p) \alpha_3 + \exp[-(2k - 1)(\sigma + Q_\sigma)/(3Q_\sigma)] = W_{N=0}(1 + f(N)),$$

with $\alpha_1 = N_1, \alpha_2 = \frac{3}{2}N_2, \alpha_3 = \frac{1}{2}N_3$. Using an exhaustion process based on $w = W_{N=0}(1 + f)$, the quantum numbers n, m, p, σ are determined via eqs. (B15)-(B21) and (B40). The steps are:

$$\begin{aligned}
w - K^3 \alpha_1 &\geq 0 \quad \text{until sign change, giving } K_n, \text{ then } w - (K_n - 1)^3 \alpha_1 = w_1, \\
w_1 - K^2 \alpha_2 &\geq 0 \quad \text{giving } K_m, \text{ then } w_1 - (K_m - 1)^2 \alpha_2 = w_2, \\
w_2 - K \alpha_3 &\geq 0 \quad \text{giving } K_p, \text{ then } w_2 - (K_p - 1) \alpha_3 = w_3, \\
w_3 - e^{-\beta K} &\leq 0 \quad \text{with } \beta = (2k - 1)/(3Q_\sigma), \text{ giving } K_\sigma.
\end{aligned}$$

Then:

$$n = K_n - 1 - Q_n, \quad m = K_m - 1 - Q_m, \quad p = K_p - 1 - Q_p, \quad \sigma = K_\sigma - 1 - Q_\sigma.$$

These are used in the mass formula (B3).

3 The Average Life Times of the Basic States

Let T be the average lifetime of particles with mass M from eq. (B3). For $T_N = T(N) \ll T$ with $T_0 = 0$, the unified relation is:

$$(T - T_N) = \frac{192hHy}{Mc^2 [\eta_{2,2}(1 - \sqrt{\eta})^2(1 - \sqrt{\eta_{1,1}})^2(1 - \sqrt{\eta_{1,2}})^2] (H + n + m + p + \sigma)(n + |m| + |p|\beta_{(0)})} \delta,$$

where $\delta = \delta(N)$ as in eq. (B7), M from eq. (B3), H from eq. (B24), and:

$$y = F \left[\phi + (-1)^s(1 + \phi) \left(b_1 + \frac{b_2}{W_{N=0}} \right) \right].$$

The expressions for $\phi, U, Z, F, s, b_1, b_2, \beta_{(0)}, D$ are given in eqs. (B49)-(B57) of the 1989 document. Lifetimes T for $N = 0$ are computed in multiples of 10^{-8} seconds.

4 The Sommerfeld Finestructure Constant

The fine-structure constant α is given by:

$$\alpha\sqrt{1-\alpha^2} = \frac{9\vartheta}{(2\pi)^5}(1-C'), \quad 1-C' = 1 - \frac{1+\eta_{2,2}}{\eta\eta_{1,1}\eta_{1,2}} \left(\frac{1-\sqrt{\eta}}{1+\sqrt{\eta}} \right)^2 = K_\alpha.$$

With $D' = (2\pi)^5/(9\vartheta K_\alpha)$, the reciprocal square solutions are:

$$\alpha_{(\pm)}^{-2} = \frac{1}{2}D'^2 \left(1 \pm \sqrt{1-4/D'^2} \right).$$

Numerically:

$$\alpha_+ = 0.72973525 \times 10^{-2}, \quad \alpha_- = 0.99998589, \quad 1/\alpha_{(+)} = 137.03601, \quad 1/\alpha_{(-)} = 1.0000142.$$

The positive branch agrees with the empirical value $1/\alpha = 137.0360114 \pm 3.4 \times 10^{-8}$.

5 The Masses of Neutrino States

Assuming a Euclidean metric in the central region (no structure element), then $L_{(n)} = -Q_n$ and $M_0 = 0$. All structure zones become Euclidean. Substituting $n = -Q_n, m = -Q_m, p = -Q_p, \sigma = -Q_\sigma$ into eq. (B3) gives $G + F + S = \phi$. Even with $\sigma + Q_\sigma = 0$, generally $\phi \neq 0$ and $\Phi \neq 0$. If $\Phi + \phi \neq 0$ (since $P > 0$ or $Q > 0$), a field mass exists, interpretable as a “spin-potence” or field catalyst—properties that necessitate neutrinos empirically.

The neutrino mass is:

$$M_\nu = \mu\alpha_+(\Phi + \phi_0),$$

where ϕ_0 is eq. (B49) evaluated at the lower bounds of n, m, p, σ . Thus M_ν depends only on k, K, P, Q . Possible neutrino states (ν_i and antiparticles $\bar{\nu}_i$) are:

$$\text{For } k = 1 : \quad \nu_1(1010), \nu_2(1110), \nu_3(1200),$$

$$\text{For } k = 2 : \quad \nu_4(2110), \nu_5(2111).$$

The β -neutrino is identified with ν_1 , the μ -neutrino with ν_2 . Masses are in eV (see Table II).

6 Concluding Remarks

For $N > 0$ states, the system (B32) is used, but uncertainties remain due to unknown $z(N)$ in $Q(N) = Q(0) + 2z(N)$. The approximation $z = 0$ yields errors below 0.1 MeV. Despite this, the calculated excitation spectra match known resonances (CERN 1973). Many theoretical terms lack empirical counterparts, possibly due to unknown selection rules or measurement limitations.

Heim intended to compute interaction cross-sections after determining $z(N)$ and T_N , but this was not completed. Nonetheless, the theory’s correspondence with data suggests it fulfills the requirements of a unified geometric-dynamical description of microphysics.